

Entangled Interlocked Diamond-like (Diamondynes) Lattices

Carlos Maciel de Oliveira Bastos^{1,2}, Emanuel José Alexandrino dos Santos¹, Rodrigo Alkimim Faria Alves¹,

Alexandre Cavalheiro Dias^{1,2}, Luiz Antonio Ribeiro Junior¹, and Douglas Soares Galvao³

1- Institute of Physics, University of Brasília, Brasília, Federal District 70919-970, Brazil.

2-International Center of Physics, University of Brasília, Brasília, Federal District 70919-970, Brazil.

3-Department of Applied Physics, State University of Campinas, Campinas, São Paulo 13083-859, Brazil.



Contact Info



Bastos, C. M. de O., dos Santos, E. J. A., Alves, R. A. F., Cavalheiro Dias, A., Ribeiro Junior, L. A., & Galvão, D. S. (2025). Entangled Interlocked Diamond-like (Diamondynes) Lattices. ACS Omega, 10(39), 46065–46070. <https://doi.org/10.1021/acsomega.5c07159>

Introduction

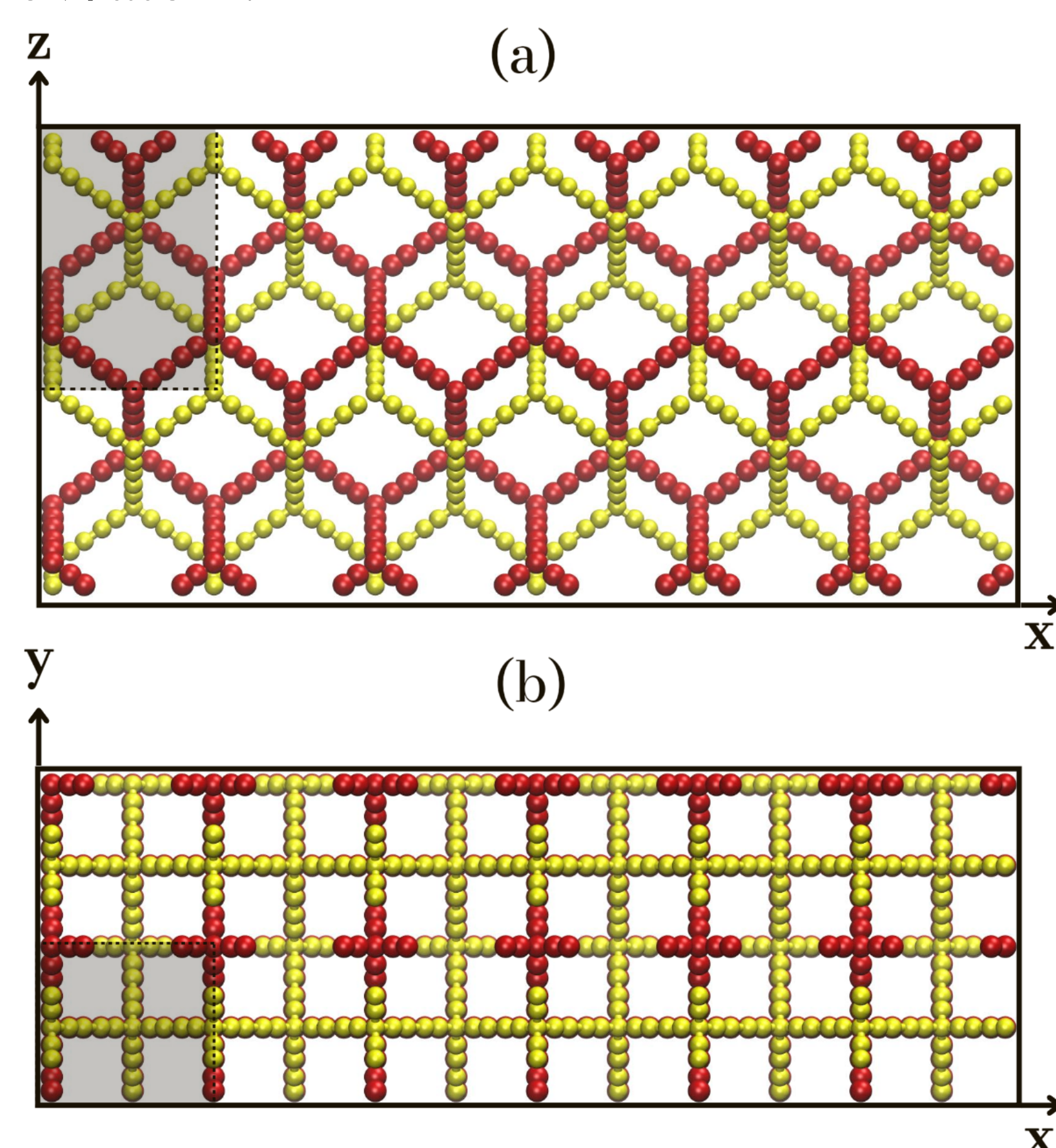
- Investigate structural and electronic properties of a new family of 3D carbon allotropes (diamondynes).
- Analyze the stability of interlocked lattices with sp-sp³ hybridization.
- Explore potential applications in energy storage and nanoscale mechanics.
- This work unveils the first diamond-like structures with entangled yet movable sublattices

Metodology

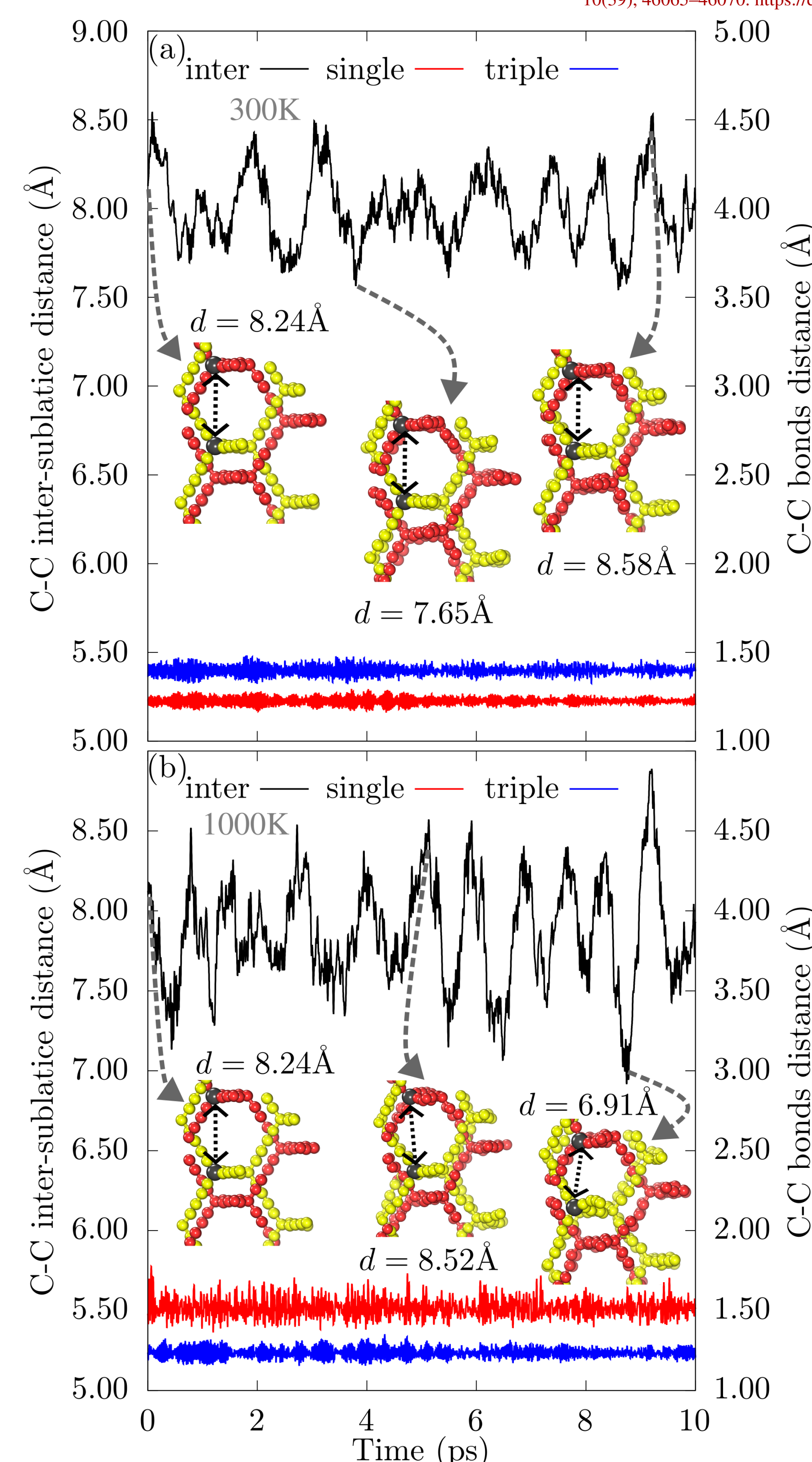
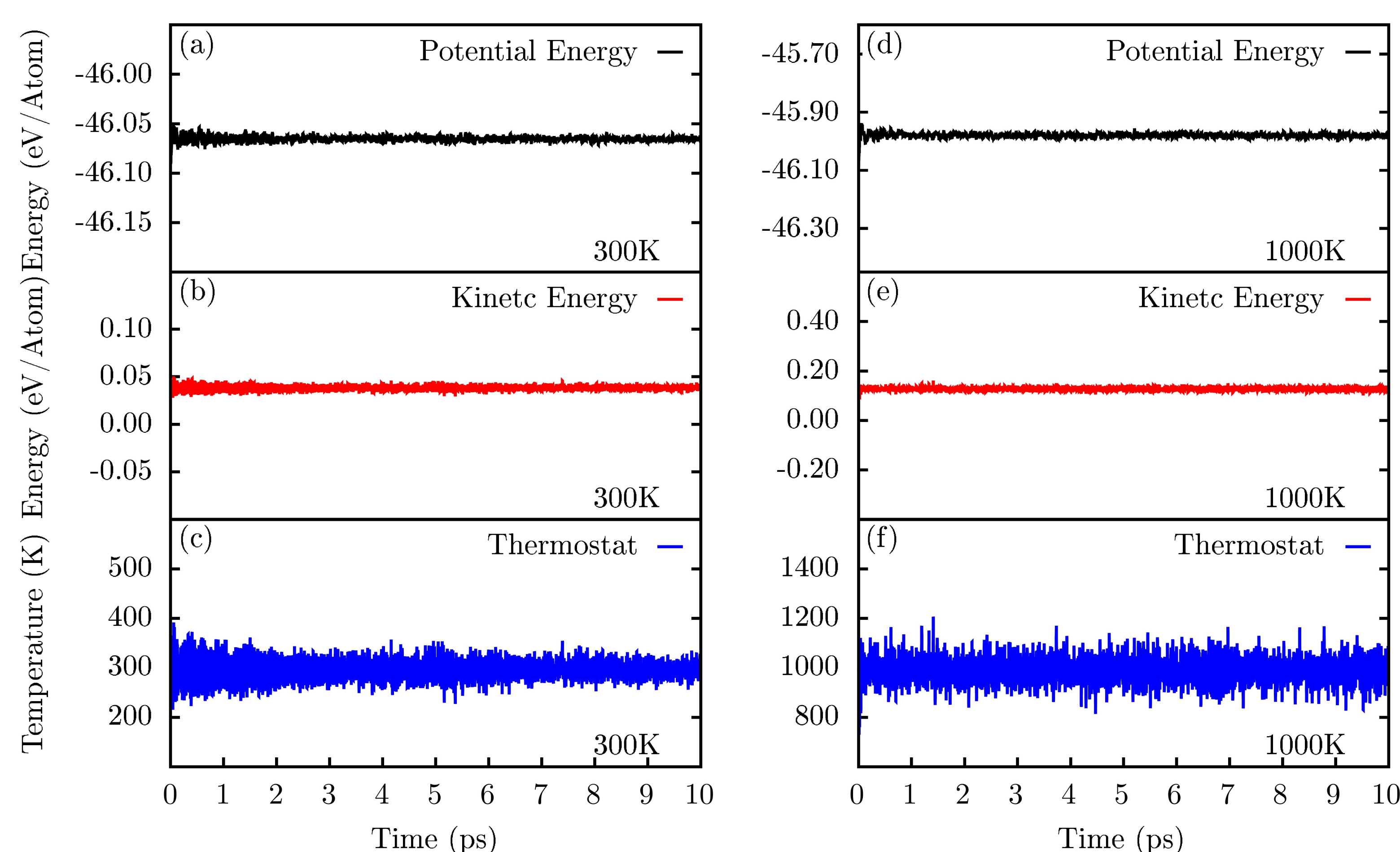
- DFT Calculations:** Performed using the SIESTA code with the PBE exchange-correlation functional and DZP basis set. HSE06 hybrid functional used for accurate electronic band gap prediction.
- Dispersion Correction:** van der Waals interactions included via Grimme's DFT-D3 method.
- Molecular Dynamics:** AIMD simulations (300 K and 1000 K) performed using the DFTB+ code to assess thermal stability.

Results

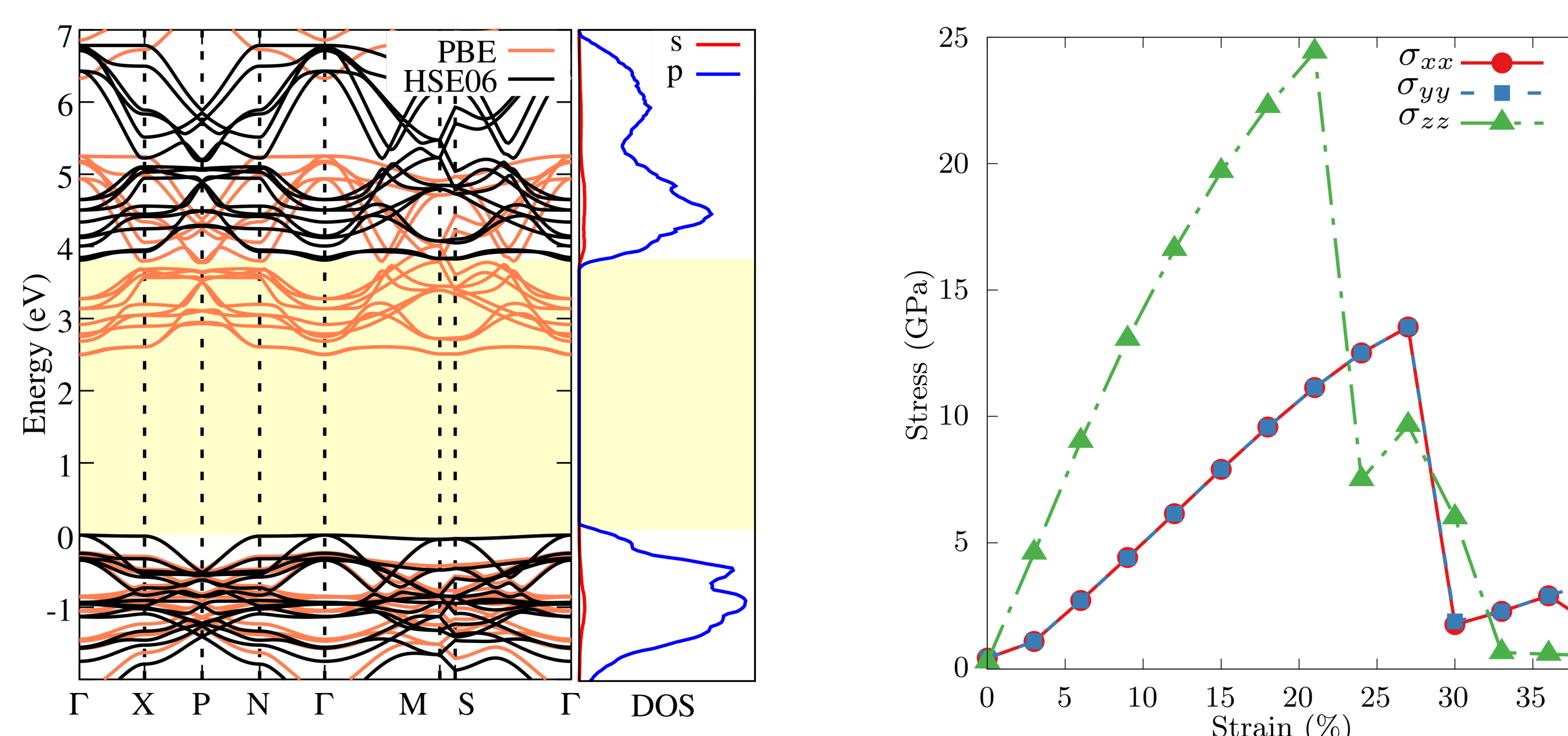
- Thermodynamic stability confirmed with cohesive energy approx. -6.7 eV/atom.



- No phase transition or structural degradation observed in AIMD simulations up to 1000 K.



- Unique relative mobility between entangled sublattices without covalent bonds.
- Wide electronic band gaps ranging from 2.2 to 4.0 eV.
- High flexibility with strain at failure around 25%
- Tunable mechanics: Young's modulus from 19 to 137 GPa.



Conclusion

- First report of crystalline, interlocked, and movable structures.
- Combines stability with flexibility and tunable electronic properties.
- Promising framework for porous energy conversion technologies.

Acknowledgments

Supported by

